

# MATHEMATICS

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## MATHEMATICS GRADE 12 TERM 3 CONTROLLED ASSESSMENT PACK

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|-----------------|-----------------------------------------------------------|
| SUBJECT         | MATHEMATICS                                               |
| GRADE           | 12                                                        |
| ASSESSMENT TASK | MATHEMATICS GRADE 12 TERM 3<br>CONTROLLED ASSESSMENT PACK |
| MARKS           | 50                                                        |
| TIME            | 1 hour                                                    |

Educator approval required before classroom use.

INSTRUCTIONS AND INFORMATION

- 1. Read all instructions carefully.
- 2. Answer all questions or complete all activities as instructed.
- 3. Show all working where calculations, planning, diagrams or explanations are required.
- 4. Use the space provided by the educator, answer booklet or learner task book.
- 5. Write neatly and clearly.
- 6. Do not use a calculator unless the educator allows it.

VISUAL MATERIAL

FIGURE 1: Working Grid And Method Marks

|                                                                    |                                                                                          |                       |  |
|--------------------------------------------------------------------|------------------------------------------------------------------------------------------|-----------------------|--|
| Mathematics                                                        |                                                                                          | CALCULATION WORKSPACE |  |
| Mathematics Grade 12 Term 3 Controlled Assessment Pack             |                                                                                          |                       |  |
| Working grid and method marks                                      |                                                                                          |                       |  |
| 1.1 (6)                                                            | Solve a routine problem linked to the stated term focus and show all algebraic working.  | <div></div>           |  |
| 1.2 (6)                                                            | Interpret the supplied graph, table or diagram and state the meaning of one key feature. | <div></div>           |  |
| 1.3 (8)                                                            | Complete a multi-step calculation using the supplied values and units.                   | <div></div>           |  |
| 2.1 (10)                                                           | Set up and solve a non-routine problem from the term context.                            | <div></div>           |  |
| Award method marks from visible working before final-answer marks. |                                                                                          |                       |  |

QUESTIONS

QUESTION 1: SECTION A: CONCEPTS AND PROCEDURES

Instructions: Show all working.

Stimulus: Topic focus: statistics, probability, Euclidean geometry and measurement.

1.1. Solve a routine problem linked to the stated term focus and show all algebraic working.  
(6)

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**1.2.** Interpret the supplied graph, table or diagram and state the meaning of one key feature. **(6)**

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**1.3.** Complete a multi-step calculation using the supplied values and units. **(8)**

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## **QUESTION 2: SECTION B: PROBLEM SOLVING**

**Instructions:** Use the workspace and justify each step.

**Stimulus:** Problem context: data displays, probability rules and geometric proof.

**2.1.** Set up and solve a non-routine problem from the term context. **(10)**

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**2.2.** Justify one method step or proof statement. **(5)**

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## **QUESTION 3: SECTION C: INTEGRATED APPLICATION**

**Instructions:** Combine concepts from the term pack.

**Stimulus:** Integrated focus: mixed-topic source-based calculation paper.

**3.1.** Answer an integrated application question and comment on the reasonableness of the result. **(15)**

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